MediRep

AI-Powered Medical Report Analyzer & Longitudinal Health Tracker

A privacy-preserving biomedical NLP system



AI Mini Project | Academic Year 2025–26

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Why Reports Confuse Patients



Clinical jargon

Reports use physician-level terms, ranges and abbreviations with no plain-language help.



AI hallucination

LLMs invent lab ranges and misread negation, such as treating “No pneumonia” as pneumonia.



Privacy risk

Sending records to cloud APIs breaks health-privacy rules such as HIPAA.

70%+ of outpatients report acute stress reading results alone, while those who dismiss abnormal values may delay care.

Project Objectives



1. Multi-format ingestion

PDF, scans and photos (up to 10 MB) read in memory with OCR



2. Deterministic lab engine

Range checks only against values printed in the report



3. Biomedical NER

8-class clinical taxonomy plus assertion classification



4. Dataset grounding

6,168 cleaned MTSamples and IU X-Ray reports



5. Longitudinal tracker

Multi-month biomarker curves with healthy baseline bands



6. Zero-retention privacy

No patient data written to disk, databases or third parties

Gaps in Existing Systems

System	Lab range accuracy	Negation handling	Privacy
Commercial LLMs	No: hallucinates	Inconsistent	Cloud upload
BioBERT	No: entities only	Needs fine-tuning	Host-dependent
ScispaCy	No: entities only	Rule add-ons	Local
Hospital portals	Partial	None	Closed system
MediRep	**Yes: strict intervals**	**Yes: 5 assertion states**	**Yes: 100% in-memory**

Only MediRep closes all three gaps: range math, negation, and privacy.

System Architecture

1 Frontend SPA

React 19, TypeScript, Tailwind; drag-and-drop upload and SVG charts

2 FastAPI server

Validates uploads, schedules background tasks, serves REST responses

3 Ingestion & OCR

PyMuPDF for digital PDFs, Tesseract for scanned reports

4 Range engine

Regex extraction and strict interval checks, no generative math

5 NER engine

Tags 8 entity classes and maps them to plain explanations

6 Trend tracker

Stores serial results and draws trajectory curves with deltas

Seven-Stage Analysis Pipeline

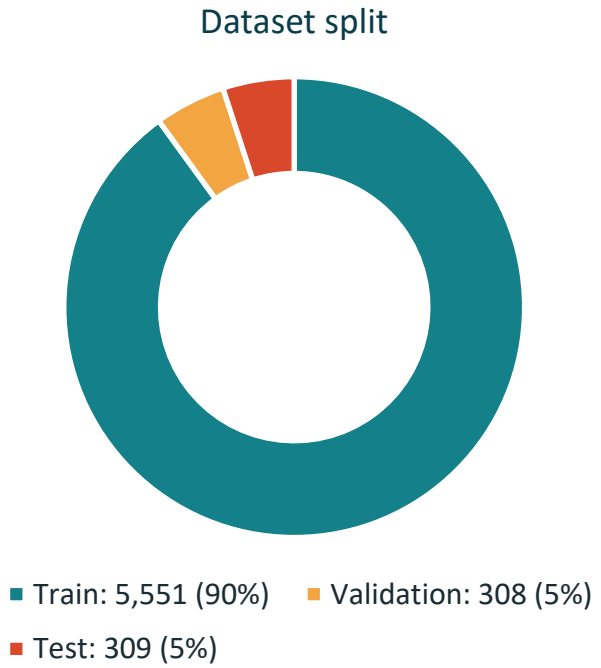
1. **Ingestion** validates type and the 10 MB limit, then PyMuPDF text or OCR
2. **Lab engine** regex pulls values and flags High, Low or Normal vs printed ranges
3. **NER** tags 8 entity classes with offsets and confidence scores
4. **Glossary** plain-language definitions at a 6th–8th grade reading level
5. **Safety** red-flag terms raise an urgent alert and pause routine tips
6. **Guidance** diet, exercise and precaution tips plus doctor-visit questions
7. **Longitudinal** numeric results update the patient's time-series curves

Dataset: 6,168 Real Reports

6,168 Cleaned, deduplicated reports in the combined corpus

2,336 MTSamples transcripts spanning 40 specialties

3,835 Indiana University chest X-ray reports (NLM/NIH)



Clinical Entity Taxonomy

DISEASE

Anemia, Pneumonia

SYMPTOM

Fever, knee pain

MEDICATION

Ibuprofen, Metformin

ANATOMY

Left knee, Lungs

TEST

Chest X-Ray, MRI

PROCEDURE

Arthroscopy, Biopsy

PATHOGEN

Streptococcus, Influenza

RADIOLOGICAL_FINDING

Pleural effusion

Assertion states

PRESENT

ABSENT

UNCERTAIN

HISTORICAL

NOT_SPECIFIED

“No pneumothorax” is tagged ABSENT

RESULTS AND OUTPUT



Future Scope & Conclusion

Voice in regional languages

Read summaries aloud in Malayalam, Hindi, Tamil, Telugu and Kannada

Chat with your report

On-device RAG assistant bound strictly to the uploaded document

Drug interaction checks

Warn about hazardous drug-drug and food-drug combinations

Conclusion

- Deterministic range checks remove numeric hallucination
- 8-class NER with negation-aware assertions
- Grounded in 6,168 clinical reports, zero data retention
- Clear, actionable insight for better doctor visits